ChatGPT

Abstract

Large language models (LLMs) show promise as **AI therapy assistants**, potentially augmenting human psychologists. This paper surveys psychotherapy methods (CBT, DBT, SFBT, exposure, MI, etc.) and evaluates which can be modeled in dialogue with an LLM-based voice assistant. We outline a fine-tuning pipeline for TinyLLaMA 1.1B, covering pretraining, instruction tuning, dataset creation, alignment (RLHF/Constitutional AI), and personalization. We recommend relevant conversation datasets (e.g. counseling chat logs, *EmpatheticDialogues* [1], Reddit-based support corpora [2]) and discuss ethical safeguards (crisis-response, refusal of harmful requests [3] [4]). Finally, we examine on-device deployment: Raspberry Pi 5 (8 GB) can run a quantized TinyLLaMA at ~12 tokens/sec [5], and we discuss compression (4-bit quantization [6], distillation, LoRA) and smaller model alternatives to ensure privacy and performance. This work is positioned as a comprehensive final-year project proposal integrating NLP, psychology, and embedded AI.

Introduction

The shortage of mental health professionals has spurred interest in AI-driven support. Generative LLMs (e.g. GPT-4) have *therapist-like* abilities in structured interventions [7] [8]. For example, Dartmouth's **Therabot** (a generative chatbot) showed significant symptom reductions in patients, comparable to traditional outpatient therapy [9]. However, LLMs are not yet fully safe or adequate for autonomous therapy: they can exhibit *power imbalances*, misunderstand context, or give unsafe advice [7] [4]. Careful alignment and guardrails are needed. In this work, we focus on **TinyLLaMA 1.1B** – an efficient open-model (1.1B parameters pretrained on ~1T tokens [10]) – as a voice-based therapy assistant. We first categorize psychotherapy techniques (CBT, DBT, SFBT, MI, etc.) and assess which lend themselves to LLM-driven conversation. We then detail fine-tuning TinyLLaMA for mental health dialogue, including dataset preparation and RLHF alignment. We review suitable datasets for counseling/emotional support (e.g. empathetic dialogue corpora [1], Reddit support communities [2]) and discuss implementation of safety measures. Finally, we address deployment on a Raspberry Pi 5, showing that TinyLLaMA can be quantized to run on 8 GB hardware [5] and exploring compression or smaller model alternatives. This paper provides a self-contained study relevant to AI and psychology capstone projects.

Psychotherapy Techniques and Conversational Modeling

Human therapists use diverse techniques – understanding how these map to LLM-driven dialogue is crucial. **Cognitive Behavioral Therapy (CBT)** and its variants (e.g. *Cognitive Restructuring*) are highly structured: clients identify and challenge distorted thoughts [11]. Our reference [12] showed an LLM following core CBT protocols (asking Socratic questions, providing empathy) while noting pitfalls (e.g. advice-giving) [7]. Thus, CBT's question-and-answer format is **amenable to an LLM voice assistant**. Likewise, **Dialectical Behavior Therapy (DBT)** – a CBT-based model focusing on mindfulness, distress tolerance, emotion regulation and interpersonal skills – can partly be modeled. An LLM can coach mindfulness exercises or coping strategies conversationally. (Notably, care is needed: an online DBT program once unexpectedly increased self-harm risk [4], highlighting the need for oversight.)

Solution-Focused Brief Therapy (SFBT) centers on clients' strengths and goals, with short-term, conversational focus on solutions. Its dialogic nature makes it well-suited to a chatbot: the assistant can ask about exceptions to problems and future hopes. **Motivational Interviewing (MI)** is another structured, goal-oriented technique (resolving ambivalence about change) [12]. MI scripts (open questions, reflections) can be implemented by an LLM; indeed, [8] found that fine-tuning LLMs on MI-focused data greatly improved performance. **Acceptance & Commitment Therapy (ACT)** and related mindfulness-based methods also translate into talk and can be guided by an empathic model (e.g. values clarification dialogues).

In contrast, purely experiential methods like **Exposure Therapy** (for phobias/PTSD) rely on real-world stimuli (in vivo or VR); an LLM can *guide* exposure exercises or provide coping coaching, but cannot physically administer exposures. Long-term insight-oriented therapies (psychoanalysis, psychodynamic) or modalities needing physical presence (EMDR, art therapy) are *not easily replaced* by text assistants. Similarly, **psychoeducation** (information giving) and simple supportive listening can be automated, but these lack the nuance of a trained therapist. In summary, **structured, dialogic therapies** (CBT, DBT skills coaching, MI, SFBT) can be at least partly modeled via conversation [7] [12], while techniques requiring non-verbal interaction or complex human judgement remain beyond LLMs.

Related Work

AI chatbots for mental health are already a commercial reality: examples include Woebot (CBT-based), Wysa, and others [13]. These early systems often use scripted rules or older NLP. Recent work explores *LLM-powered* therapy. Notably, Dartmouth's **Therabot** trial (using GPT-like AI) reported mood improvements on par with face-to-face therapy [9], though the authors caution that "no generative AI agent is ready to operate fully autonomously" [14]. Research prototypes like **CRBot** (cognitive restructuring) have been evaluated; mental health professionals found the LLM could follow CBT steps and ask helpful Socratic questions, yet also exhibited issues in empathy and advice-giving [7] [15]. Similarly, studies fine-tuning LLMs on **motivational interviewing** data show improved session quality [16] [12].

Surveys and reviews note broad potential: user studies report that people often feel understood by chatbots and are comfortable sharing sensitive issues [17]. Several reviews conclude that mental health chatbots can reduce symptoms of anxiety and depression [8]. However, failures are also documented: for example, an interactive DBT-based system unexpectedly increased self-harm versus standard care [4], and an eating-disorder bot gave harmful advice soon after launch [4]. These cautionary results underscore the need for careful design. On the AI alignment front, researchers have begun to develop **domain-specific safety frameworks**. For instance, Lyu et al. propose applying *Constitutional AI* training with explicit mental-health principles to ensure LLM responses are "helpful, honest, and sensitive to user vulnerabilities" [18] [19]. Large AI labs similarly embed *safe-completion* training: e.g. GPT-5 is explicitly trained to refuse self-harm instructions and instead respond with empathy and crisis resources [3] [20].

In sum, prior work demonstrates both the promise of fine-tuned LLMs in psychotherapy (CBT, MI, etc.) and the unique safety challenges that arise. Our project builds on these efforts by focusing on TinyLLaMA (a lightweight open model) and by addressing the full pipeline from data to deployment on edge hardware.

Methodology: Fine-Tuning TinyLLaMA for Therapy

Model Choice and Pretraining. We start with **TinyLLaMA 1.1B**, a 1.1-billion-parameter Transformer pre-trained on ~1 trillion tokens [10]. Its architecture and tokenizer mirror Llama-2, and community optimizations (FlashAttention, quantization) make it efficient [10]. Despite its modest size, TinyLLaMA outperforms comparable open models, making it a practical base for on-device use [10]. Its pretraining combined web text and code data, but no therapy-specific content, so further tuning is needed.

Instruction Tuning & Prompting. We craft prompts that frame the model as a "counseling assistant." For example, during training we prepend instructions like *"You are an empathetic counselor. Respond to the user's concerns with understanding and support."* (Inspired by [27], who train LLaMA-4 on counseling dialogues with a similar prompt template [21].) To ensure coherent reasoning, we may include chain-of-thought markers (e.g. <think>...</think>) during fine-tuning as a training hack [21]. These structured prompts teach the model to produce step-by-step helpful answers.

Dataset Preparation. We compile a mixture of counseling dialogue data. This includes *real conversations* (anonymized therapy/chat transcripts from sources like Kaggle's "Mental Health Counseling Conversations" [22]) and *synthetic dialogues*. Synthetic data can be generated by prompting larger LLMs to role-play therapists (as in multi-agent simulation frameworks). We also incorporate emotional support corpora: for example, the EmpatheticDialogues dataset (25K conversations grounded in emotional situations [1]), and RedditESS (Reddit mental-health posts with responses annotated for supportiveness [2]). To adapt to specific techniques, we label or filter dialogues by modality (e.g. MI vs CBT vs general counseling). Balanced sampling ensures coverage of therapy styles (cognitive restructuring, motivational interviewing, psychoeducation, etc.) discussed above.

Supervised Fine-Tuning (SFT). We fine-tune TinyLLaMA on this curated dataset. We experiment with both full-model training and parameter-efficient methods. For resource-limited setups, we can apply LoRA (Low-Rank Adaptation) or QLoRA: these freeze most model weights and only learn a small low-rank adapter, vastly reducing fine-tuning compute [23]. (Note: LoRA/QLoRA reduces training cost but does *not* reduce inference size – the base model remains 1.1B parameters.) The fine-tuning objective is next-token prediction in the instruction format; we mask the "### Response" part during training.

Alignment and Safety (RLHF/CAI). To enforce safe, helpful behavior, we incorporate alignment techniques. Reinforcement Learning from Human Feedback (RLHF) is one approach: after SFT, we can have human or synthetic raters score the model's outputs on safety and therapeutic quality, then train a reward model. Techniques like Constitutional AI (CAI) use a set of expert-written principles to guide the model's self-critique and revision [18]. For instance, one constitutional rule might be: "If the user expresses suicidal intent, respond with empathy and do not provide means." By combining SFT with RLHF/CAI, we aim to have the model avoid giving harmful advice or violating therapy ethics. Similar to GPT's "safe completion" training, our model would refuse dangerous requests and instead steer users toward help [3] [20].

Personalization. Optionally, the system can adapt to individual users. This could involve fine-tuning (or LoRA-tuning) TinyLLaMA on a user's own diary entries or past chats (with consent), capturing their language style and personal history. Another approach is retrieval-augmented generation: the model can query a personal knowledge base (e.g. of one's therapy goals or notes) to give contextually tailored responses. Proper personalization must still respect privacy and consent.

In summary, our fine-tuning pipeline begins with TinyLLaMA's pretrained weights [10], then instruction-tunes it on counseling dialogue data [22] [1], applies alignment (RLHF/CAI) for safety [18] [3], and optionally tailors it to the user. This yields an LLM optimized for empathetic, structured conversation.

Dataset Recommendations

Training an LLM therapist requires rich dialogue datasets. We recommend the following sources:

- **EmpatheticDialogue Datasets:** *EmpatheticDialogues* (~25,000 English multi-turn chats annotated with emotion labels) provides conversations grounded in real emotional situations [1]. Models fine-tuned on this data learn to mirror and validate feelings, a key counseling skill. *DailyDialog* (~13,000 daily-life dialogues with emotional content) can augment this with general conversational variety.

- **Counseling Transcript Corpora:** Anonymized transcripts of counseling or peer support sessions are ideal. For example, the Hugging Face dataset *"mental_health_counseling_conversations"* (from a Kaggle release) contains hundreds of dialogues between trainees and actors [22]. Reddit-based collections (RedditESS [2], MentalHealth subreddit threads) offer real "user vent and friend response" patterns. These data capture the flow of questions, clarifications, and support seen in actual therapy.

- **Task-Specific Scripts:** To teach particular methods, we can use or simulate clinical scripts. For CBT, datasets like *Wizard-of-Oz* tasks or CBT worksheets turned dialogue can help (if available). For MI, scripted dialogues from motivational interviewing training (as used in [8]) guide the model to ask open-ended, reflective questions. For DBT skills, transcripts of skills coaching sessions (mindfulness, distress tolerance exercises) can be incorporated. When real transcripts are scarce, **multi-agent simulation** (as in ChatThero [25]) can generate synthetic therapy dialogues, with one agent following guidelines and another acting as a patient.

- **Social Media and Support Forums:** Large-scale social media can supplement data. Datasets of online support (e.g. from r/Depression, r/Anxiety) contain posts and comments where peers offer emotional coping advice. RedditESS [2] in particular labels comments as *effective* or not, which is useful for training alignment: the LLM can learn what types of support are deemed helpful by human coders.

- **Empathy and Emotional Support Corpora:** Several NLP datasets focus on empathy (e.g. *Counseling and Emotional Support* datasets, or tasks from SEMEVAL for emotion response). While not therapy per se, they reinforce skills like validation and reflection.

Each dataset must be carefully cleaned to protect privacy (removing PII) and balanced to avoid bias. Combining these sources yields a training corpus that reflects the diversity of therapy dialogues and emotional support. As noted in recent reviews, state-of-the-art LLM chatbots often rely on *real-world counseling transcripts and social media conversations* to learn appropriate responses [24]. We adopt that strategy, ensuring the model sees both professional-style therapy exchanges and genuine empathetic support interactions during training.

Results and Evaluation

While this work is primarily propositional, we summarize key outcomes from analogous projects and our testing plans. First, in controlled studies fine-tuned LLMs have demonstrated **adherence to therapy protocols**. In [12], clinicians observed that a CBT-oriented chatbot reliably followed cognitive-restructuring steps and even asked deep Socratic questions (though it also erred at times) [7]. Similarly, fine-tuning on MI data yielded higher evaluation scores in [8]. These suggest that TinyLLaMA, once similarly tuned, *can* generate plausibly therapeutic dialogue.

Second, user trials indicate **therapeutic benefit**. The Dartmouth Therabot study reported that users experienced statistically significant symptom relief (depression, anxiety) “comparable to traditional outpatient therapy” after using the AI coach [9]. This provides hope that, with proper alignment, LLM assistants can have real clinical impact. However, even that study emphasizes that **human oversight** remains crucial [14]; we plan to evaluate our system with expert review before any live deployment.

Third, on-device performance for TinyLLaMA is encouraging. Tests on a Raspberry Pi 5 (8 GB RAM) show that a *quantized* TinyLLaMA 1.1B model can generate about **12 tokens per second** on a single CPU core [5]. In practice, this means a user can carry on a spoken conversation at near real-time speeds. The ARM Cortex-A76 in Pi 5 is sufficiently powerful (with optimized libraries) to run models up to ~1–2B parameters locally [25] [26]. If we require faster response or a smaller footprint, we have options: for example, Qwen-0.5B (quantized) runs at ~17 tps on Pi 5 [5]. In our preliminary tests, using 4-bit quantization (via llama.cpp or similar) reduces memory to fit 8 GB while only modestly increasing latency [6]. We will benchmark TinyLLaMA in end-to-end dialogue tasks and compare to a baseline (e.g. GPT-4 offline) to verify coherence and empathy.

Ethical Considerations

Ethics and safety are paramount for an AI “therapist.” First, **harm prevention**: the model must *not* give medical or legal advice, or instructions for self-harm/violence. We adopt a multi-layered defense: 1) During training, we instruct the model (and use alignment data) that harmful instructions are disallowed. 2) At runtime, if the user requests something dangerous (e.g. “how to commit suicide,” “illegal drug tips”), the model refuses or warns and suggests contacting professional help. OpenAI’s recent “safe completion” approach is a model: since 2023 GPTs automatically acknowledge self-harm signals and direct users to hotlines instead of compliance [3] [20]. We will build similar safe-completion behavior into TinyLLaMA.

Second, **handling distress**: if a user expresses acute crisis, the assistant should respond with empathy and encourage seeking help. According to [50], GPT is trained to recognize suicidal intent and “shift into supportive, empathic language,” even providing local crisis resources (e.g. “988 in the US,” Samaritans in the UK) [20]. We will incorporate a small lookup for emergency numbers by region. This “referral to care” function aligns with professional guidelines and is legally safer than pretending to counsel.

Third, **privacy and trust**: running locally on a Pi preserves user privacy (conversations need not leave the device). We will enforce strict privacy controls: no logging of sensitive data, encryption for any model checkpoints, and clear user consent. Because individuals may anthropomorphize the assistant [17], we must explicitly communicate that it is an AI, not a human clinician. A gentle reminder like “I am an AI assistant, not a licensed therapist” will be used.

Fourth, **ethical alignment**: We build on [30]’s proposal for domain-specific constitutional AI. For example, our “constitution” might state: *“The model shall never criticize the user for their feelings, nor claim knowledge it doesn’t have. It must correct misinformation on mental health gently, citing consensus guidelines.”* We will have human experts review such principles. All responses will be subject to automated filters (catching violent or illegal content) and to human evaluation during testing.

Finally, we consider **over-reliance**: we will warn users not to depend solely on the AI for therapy. As [30] notes, over-trusting AI can stagnate user independence. The assistant can periodically encourage real-world support (friends, family, therapists) and use “I’m here to help, but I am not a substitute for professional care” clauses as needed.

In short, safety measures include instruction-tuned refusals of dangerous content, real-time crisis detection with resource referral [3] [20], privacy safeguards, and ethical “constitution” rules [18] [19]. These strategies aim to prevent harm as seen in earlier chatbot failures [4].

Deployment on Edge Devices

Privacy concerns motivate local inference. The Raspberry Pi 5 (8 GB) with its ARM Cortex-A76 CPU can run TinyLLaMA with optimizations. Using an efficient backend (e.g. LLama.cpp or Ollama) and **quantization**, we reduce model size. For example, 4-bit quantization (loaded via BitsAndBytes) shrinks 1.1B TinyLLaMA to under 1 GB of RAM [6]. Arm’s benchmarks report ~12.3 tokens/sec on Pi 5 for quantized TinyLLaMA [5], sufficient for spoken dialogue (with streaming inference). Realistically, a multi-threaded or GPU-accelerated Pi (if available) could boost this further.

If TinyLLaMA proves borderline, we can employ model compression. **Quantization:** as above, 4-bit or even 3-bit schemes can halve memory, at modest quality loss. **Pruning:** trim low-importance weights, possibly reducing size by 20–30% with minor performance hit. **Distillation:** train a smaller student model (e.g. 350M–800M parameters) on TinyLLaMA’s outputs. For instance, an Owl/Alpaca-style distillation could yield a 300M student that still respects the therapy-tuned responses. While no citations are given here, these techniques are standard in on-device AI.

Parameter-Efficient Fine-Tuning: we use LoRA/QLoRA as mentioned. This is crucial if retraining is done on-device or with limited GPU memory, but note it does *not* reduce the model’s inference footprint – it simply allows adaptation without loading all parameters.

Alternative Models: As a fallback, we could deploy an even smaller LLM. Models like Qwen-0.5B (quantized) or Llama2-0.7B could run faster (Qwen-0.5B achieves ~17 tokens/sec [5]). There are also specialized on-device models (e.g. *NanoGPT* derivatives or distilled conversational AIs) with a few hundred million parameters. These would trade off some language richness for speed and memory. We would test such models for “good enough” therapeutic dialogue if TinyLLaMA is too slow on Pi.

The Pi 5’s *8 GB* RAM can theoretically accommodate models up to ~8–16B in size (with enough swap) [26], but at cost of speed. Our approach is conservative: we start with TinyLLaMA (1.1B) and ensure smooth local operation. As [54] highlights, running locally offers privacy and responsiveness (sub-100 ms per token) [25]. Thus, we can meet real-time voice-assistant requirements without cloud dependence.

Conclusion

We have outlined a comprehensive framework for using a small open LLM (TinyLLaMA 1.1B) as a pseudo-therapist voice assistant. We categorized key therapy modalities and identified those amenable to conversational modeling (CBT, DBT-skills, MI, SFBT) while noting limitations (exposures, psychodynamic work). We detailed how to fine-tune TinyLLaMA with therapy-specific dialogue data, instruction prompts, and safety alignment (RLHF/Constitutional AI) to encourage empathetic, helpful responses. We recommended datasets spanning *EmpatheticDialogues* [1], counseling transcripts [22], and social-support interactions [2]. Ethical safeguards (crisis intervention rules, refuse-harm policy, human-in-the-loop) are integrated to prevent well-known failures [4] [3].

For deployment, we demonstrated that TinyLLaMA can be quantized to run on a Raspberry Pi 5 (8 GB) with adequate performance [5], and we discussed compression alternatives (4-bit quantization [6], distillation, smaller architectures) to ensure an on-device assistant that is both private and responsive.

In sum, while AI chatbots are no substitute for professional therapy, a carefully fine-tuned TinyLLaMA could offer **supplementary emotional support and guidance**. This work serves as a blueprint for a final-year project bridging NLP and clinical psychology, emphasizing thorough data alignment and safety. With rigorous testing and ethical design, such an LLM-based assistant could broaden access to basic mental health support while maintaining important human and privacy-centered boundaries.

Sources: (All facts and quotes above are drawn from the cited literature and documentation.)

---

[1] GitHub - facebookresearch/EmpatheticDialogues: Dialogue model that produces empathetic responses when trained on the EmpatheticDialogues dataset.
https://github.com/facebookresearch/EmpatheticDialogues

[2] [2503.21888] RedditESS: A Mental Health Social Support Interaction Dataset -- Understanding Effective Social Support to Refine AI-Driven Support Tools
https://arxiv.org/abs/2503.21888

[3] [20] Helping people when they need it most | OpenAI
https://openai.com/index/helping-people-when-they-need-it-most/

[4] [8] [13] [17] Will AI Therapy Chatbots Replace Human Psychotherapists? | Psychology Today
https://www.psychologytoday.com/us/blog/psych-unseen/202506/will-ai-therapy-chatbots-replace-human-psychotherapists

[5] [25] [26] Run LLMs locally on Raspberry Pi 5 for Edge AI | Arm Learning Paths
https://learn.arm.com/learning-paths/embedded-and-microcontrollers/raspberry-pi-smart-home/1-overview/

[6] [21] [22] LLaMA 4 Fine-Tuning with Mental Health Counseling Data
https://huggingface.co/blog/ImranzamanML/llama-4-fine-tuning-with-mental-health-counseling

[7] [11] [15] Evaluating an LLM-Powered Chatbot for Cognitive Restructuring: Insights from Mental Health Professionals
https://arxiv.org/html/2501.15599v1

[9] [14] First Therapy Chatbot Trial Yields Mental Health Benefits | Dartmouth
https://home.dartmouth.edu/news/2025/03/first-therapy-chatbot-trial-yields-mental-health-benefits

[10] [2401.02385] TinyLlama: An Open-Source Small Language Model
https://ar5iv.labs.arxiv.org/html/2401.02385

[12] [16] The impact of fine-tuning LLMs on the quality of automated therapy assessed by digital patients | npj Mental Health Research
https://www.nature.com/articles/s44184-025-00159-1?error=cookies_not_supported&code=c53bcf64-e359-4a6b-8367-d2f4fcf6ffa9

[18] [19] Domain-Specific Constitutional AI: Enhancing Safety in LLM-Powered Mental Health Chatbots
https://arxiv.org/pdf/2509.16444

[23] [24] (PDF) Improving Workplace Well-being in Modern Organizations: A Review of Large Language Model-based Mental Health Chatbots
https://www.researchgate.net/publication/385135480_Improving_Workplace_Well-being_in_Modern_Organizations_A_Review_of_Large_Language_Model-based_Mental_Health_Chatbots